30/09/2026

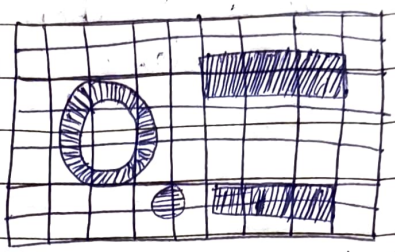
Post-Midsem

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Image Segmentation v/s classification

Image classification:-

→ It assigns a single class label to an entire image. (classify)



→ Every pixel in the scene is assigned either of the class sets.

→ concept of image slicing and bringing a binary threshold.

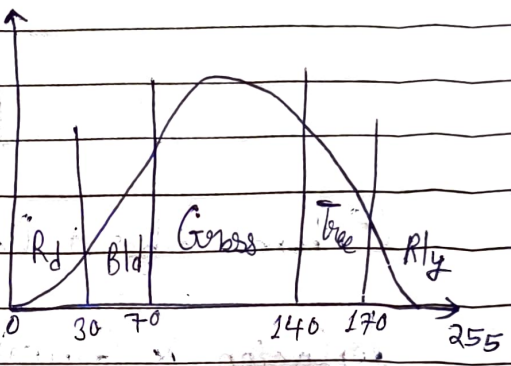


Image Segmentation:

→ It classifies visual data at the pixel level to map exact shapes and boundaries.

R_d → Road
B_d → Building
Rly → Railway track

problems during classification arise when:-

→ perfect partition of intensities among classes might not exist (significant overlapping).

→ What if the same pixel contains two or more objects.

1, 2, 3, 4 — class 1

9, 10 — class 2

17, 18, 20, 24 — class 3

K-means: (K mean saturates when means becomes constant) (K-Means is unsupervised)

02/10/2026

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Mine Sweeper and Paint Bucket: region growing

4 neighbours is less computationally expensive (at a time) as compared to 8 neighbours

2	3	1	60	70	80
5	4	2	90	100	110
1	2	3	112	113	111
5	4	4	115	85	90
3	2	5	70	80	95
1	4	6	60	90	110

if $(I - I_{seed}) < I_{threshold}$

True (same class)

else

Different class

Mean value method of Kernel Region Growing

	x	
x	seed	x
	x	

$\frac{\sum (I)}{n}$ = Seed maximum intensity (coordinates)

Region growing also gives boundaries of object

∴ Segmentation can identify multiple instances of the same class

(contours and geometry can be traced)

Depending only on brightness for classification is dangerous and the entire region will be classified as a single class.

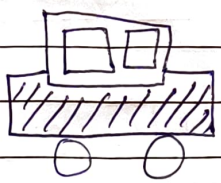
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Flood Filling Algorithm:-

0	0	0	0	0	0
0	1	1	1	1	0
0	1	0	0	1	0
0	1	0	0	1	0
0	1	1	1	1	0
0	0	0	0	0	0

put background = visited & then insert the remaining 0s

note:- Seed expansion is unable to fill up the window (∵ Holes were created)



Morphological Operator:

- Dilation
- Erosion

These are shape-based image processing techniques using a small matrix (kernel/structural element) to probe, analyze, and modify the geometric structures of objects within an image.

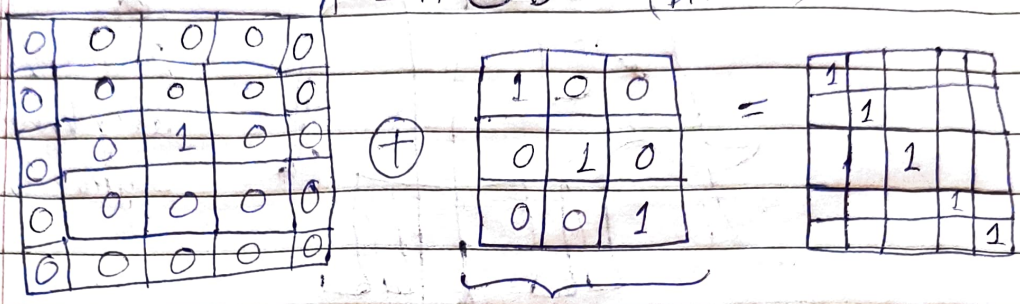
1. **Dilation:** Enlarges Expands or thickens the boundaries of a foreground objects by setting up the output pixel to the neighbourhood maximum (adds pixels where

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the kernel hits the foreground).

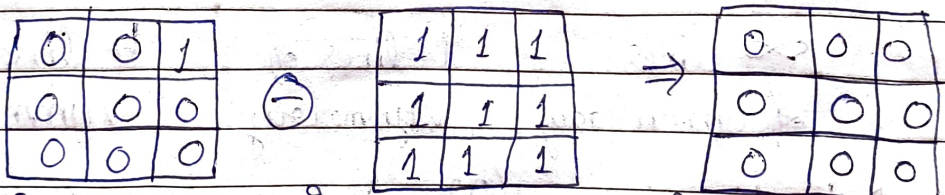
Kernel = Structuring Element

$A' = A \oplus B$ (Dilation)



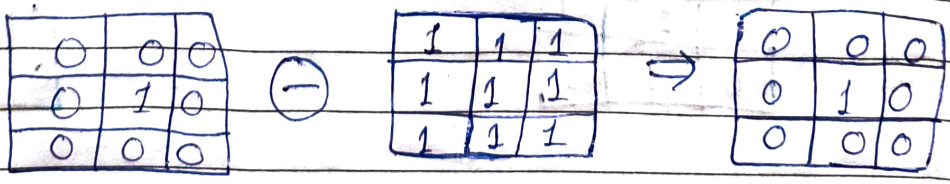
Kernel

ii. Erosion: Shrinks or thins an object boundaries by setting the output pixel to the neighbourhood minimum (retains pixels only when the kernel fits entirely inside the foreground).



Erosion अपने neighbours को खा जाता है।

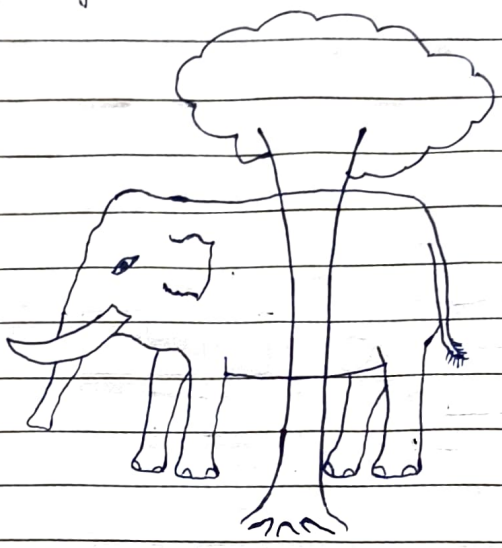
note:- Dilation can also fill up the windows but this will be more computationally expensive as compared to flood fill.



08/10/2006

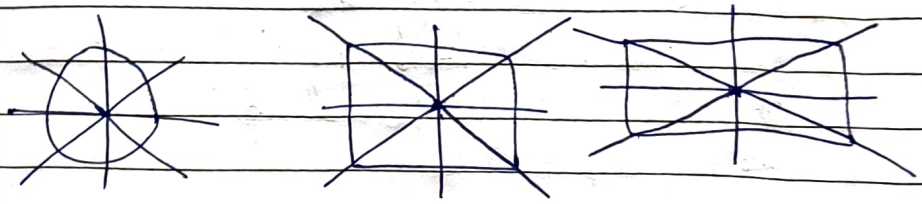
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Image patches connected via Transformers



2 instances of trees & 2 instances of elephant via segmentation

2 interval 2.



∞ lines of symmetry | 4 lines of symmetry | 4 lines of symmetry

Dilation (Morphological operator)

1.

0	0*	0
0	0	1
0	1	1

+

1	*	1

=

0	0	0
0	0	1
0	1	1

2.

0	0	0
0	1*	1
0	1	1

+

1	*	1

⇒

0	0	0
1	1	1
0	1	1

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3.

0	0	0
0	1	0
0	0	0

+

1	1	1
1	1	1
1	1	1

⇒

1	1	1
1	1	1
1	1	1

Erosion:-

0	0	0	0	0
0	1	1	1	0
0	1	1	1	0
0	1	1	1	0
0	0	0	0	0

→

0	0	0	0	0
0	0	0	0	0
0	0	1	0	0
0	0	0	0	0
0	0	0	0	0

Opening operator: Removes protrusions/noise.

$$A \circ B = (A \ominus B) \oplus B$$

⊖: erosion ⊕: dilation

Derived operator → Closing operator:

Fills small holes and small gaps

$$H * B = (H \oplus B) \ominus B$$

Boundary extraction: Original object minus its eroded version leaves the boundary

$$\beta(A) = A - (A \ominus B)$$

Hole filling operator

8 connectivity

1	1	1
1	1	1
1	1	1

4 connectivity

	1	
1	1	1
	1	

if centre pixel is 1 and within the viewing window; any 1 is present, that 1 will be categorized as its neighbour

Ques.

1	0	0	0	0
0	1	0	0	0
0	0	1	0	0
0	0	0	0	0

How many groups according to 4 connectivity?

⇒ (3)

How many according to 8 connectivity?

⇒ (1)

→ Noise removed via thresholding

→ Holes/highlights extracted via thresholding

Double Thresholding

R	R	B
R	B	B

$\leq 30 - 10$

$\geq 200 + 10$

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Transformer Encoder Computation

Split the sentence into tokens (words). Each token has a position index.

$$X = E + P$$

X token encoding - relevant embedding input representation
E : positional encoding
P : positional encoding to the encoder

Each token is mapped to a token embedding vector. A positional encoding vector is added to inject the token position information.

Check GRAMMAR, check redundancy, spellings, etc.

HEADS in vision :- brightness, no. of objects, fore-ground, background (global features)

$$\text{Attention} = \text{softmax}\left(\frac{QK}{\sqrt{d_k}}\right)V$$

Query, Key & Value

Combined = Embedding + Position Info Embedding

15/10/2026

Text-based transformer : GPT

Attention mechanisms

Attention Mechanism: It allows models to dynamically weigh the importance of different tokens in a sequence so that words capture

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rich, context-dependent meanings.

Vaswani et. al. (2017)
"Attention is All You Need"

→ causal relationships & masking effect

→ positional encoding + value embedding

→ Query (Q), Keys (K), Values (V)

$$\begin{cases} Q = XW_Q + b_Q \\ K = XW_K + b_K \\ V = XW_V + b_V \end{cases}$$

$$M_{ij} = \begin{cases} 0, & j \le i \\ -\infty, & j > i \end{cases}$$

{Applied Attention Score}

$$A_{ij} = \frac{e^{S'_{ij}}}{\sum_k e^{S'_{ij}}}$$

→ Softmax is used

→ 10 × 10 causal attention heat map generation using attention scores, softmax, and causal masking

→ The attention heatmap alone requires another operation:

$$Z = AV$$

↓

Attention weighted scores (Vector)

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→ for each query, we repeatedly calculate weighted-value vectors.

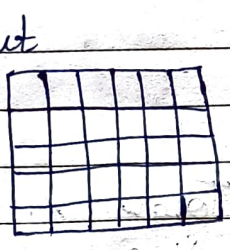
→ Vocabulary projection based on "candidate" vocabulary weight matrix estimation and next-token distribution

→ CNN vs. RNN fundamentals ✓✓

21/10/2026

Lab 10 Highlights

Input









Repository

$\frac{224}{16} = 14$

$14 \times 14 = 196 + 1 = 197$

Encoder is bidirectional meaning every word is equally related to another (in a two-way connectivity)

Decoder has a causal (masking) relation with the future words

The apple is red it is good for health

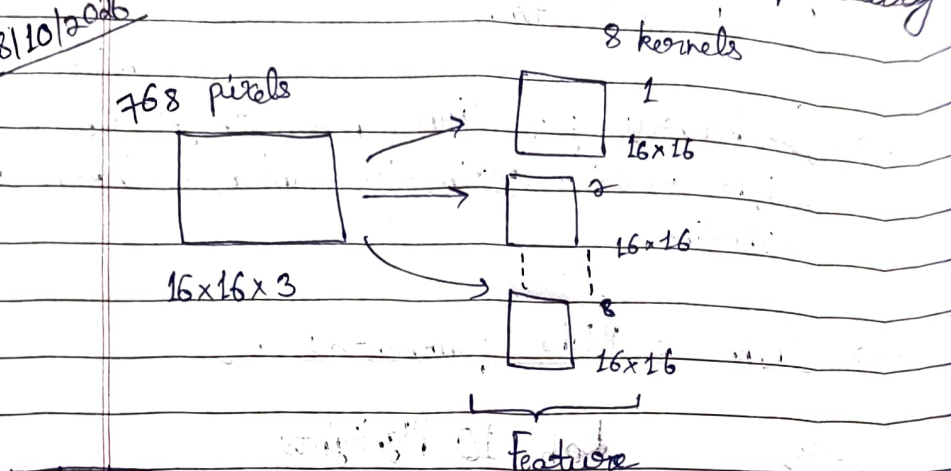
red → [0.7]
and → [0.2]
fruit → [0.1]
[0.05] tasty

done by encoder

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Poor prompts lead to hallucination possibility

28/10/2026





$w_i \; x_i \quad (1 \text{ to } 768)$

Total weights
$\{w_1, w_2, w_3, \ldots, w_8\}$

Total x_i
$\{1, 2, 3, \ldots, 768\}$

Total bias:
$\{b_1, b_2, \ldots, b_8\}$

$ReLU = \max(x, 0)$ ← हर sum (sum) के लिए एक bias

in latent space;

	1	2	3	4	5	6	7	8
1	0.2	0.4	0.6	0.8	0.2	0.1	0.9	
2								
3								
4								
5								
6								
7								
8								

} use it to find μ and σ

256 → 8 के कारण

due to presence of 8 different layers

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- In VAE, every value is calculated with a probability.
- Finding value of each pixel and the probability of its occurrence in VAE.
- FCNN: Where all elements are connected (taken as input)
  FCNN follows Multilayer Perceptron
- FCNN & ~~even~~ Transformers में forward propagation is used.
- Embedding characteristics must follow forward propagation (in transformers also)
- इसी में at decoder, unknown value is tried to be predicted. Here too, forward propagation is needed.
- ∴ We get various values ⇒ we can get μ, σ and create a VAE.

FCNN

↓

Fully Connected Neural Network

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Variational Autoencoder (VAE)

Encoder (CNN) $q(z|x)$

MaxPool → Conv2 → Max Pool → Flatten

Image

Latent distribution (reparameterization)

Feature space representation as a scatter plot

$z = \mu + \sigma \odot \underbrace{\varepsilon}_{\text{noise}}$

$\varepsilon \sim N(0, I)$

Decoder (Transposed Conv. / upSampling) $p(x|z)$

(FC → Reshape → upConv1 → upConv2 → Conv3)

Reconstructed Image

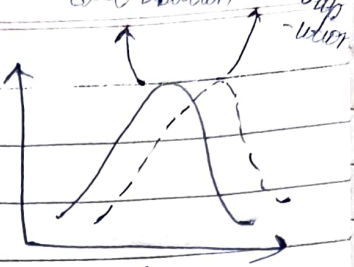
$$\mu = \begin{cases} (w_i x_i)^{set} + b_{set} \\ (16389\ w_i x_i) + b_{set} \end{cases} \Rightarrow \underbrace{(w_i, x_i)_{set}}_{\sum_{set}=1} + b_{set}$$

$64 \times 64 \times 3 = 12,288$

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$$L_{VAE} = BCE + \beta \cdot D_{KL}$$

$$= L_{Reconstruct} + \beta \cdot D_{KL}$$



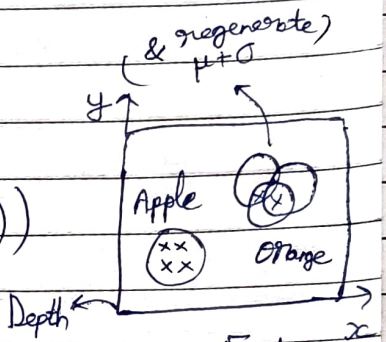
$$\text{Cross-Entropy} = \text{Entropy} + D_{KL}$$

(Concept of KL Divergence)

Covariance :-

$$\sum (H_k - X_k)(H_G - X_G)$$

$$D_{KL}(P||Q) = \sum (P(y) \log \left(\frac{P}{Q}\right))$$



Cross-Entropy:

$$- \sum (P(y) \log_2 (Q(y))$$

Q) $P_{true} = [0.8 \quad 0.2]$

$Q_{model} = [0.69 \quad 0.31]$

apple orange

Soln. $\text{Cross-Entropy} = - \sum p \log q$

$$= [-0.8 \log_2 0.69 + 0.2 \log_2 0.31]$$

$$= -[-0.42826 + (-0.337)]$$

$$= -[-0.76619] = 0.76619$$

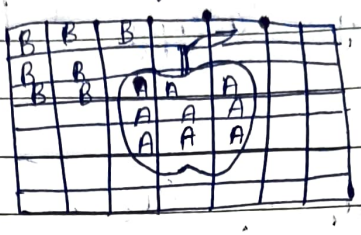
→ $$\mu = \sum_{set=1}^{8} (w_i, x_i)_{set} + b_{set}$$

04/01/2026

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Handcrafted Feature Extraction

Let 50 pixels be classified as apple & rest be classified as background



Transformer can tell which fruit is where, how it is delineated, and how others are background region

05/11/2026

HANDCRAFTED FEATURE EXTRACTION is a DIP technique where we manually design and extract meaningful features from an image instead of a machine to learn all features automatically.

Example Features:

1. Colour
   - RGB Colour Model
   - Colour Feature Vector
   - Color Histogram
2. Texture Feature extraction
   - GLCM (Gray level Co-occurence Matrix)

Homogeneity $= \sum_i \sum_j \frac{P(i,j)}{1+|i-j|}$

Contrast $= \sum_i \sum_j (i-j)^2 P(i,j)$

Energy $= \sum_i \sum_j P(i,j)^2$

Correlation $= \sum_i \sum_j \frac{(i-\mu_i)(j-\mu_j)P(i,j)}{\sigma_i \sigma_j}$

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(3.) Edge / Shape Features

→ Area = $\sum_{x,y} B(x,y)$

→ Perimeter

→ Circularity = $\frac{4\pi A}{P^2}$

→ Eccentricity = $e = \sqrt{1 - \frac{b^2}{a^2}}$

"Hand crafted Discriminator"

Histogram Apple = 50, 100, 200, 50
Histogram Banana = 20, 80, 200, 100

(Total) Test Histogram = 30, 70, 190, 88

$\Delta_1 = \sqrt{(20)^2 + (30)^2 + (10)^2 + (33)^2}$

$\Delta_2 = \sqrt{(10)^2 + (10)^2 + (10)^2 + (12)^2}$

and now classify w.r.t. least distance

GLCM Feature Formulas

1. Contrast = $\sum_i \sum_j (i-j)^2 P(i,j)$

2. Energy = $\sum_i \sum_j P(i,j)^2$

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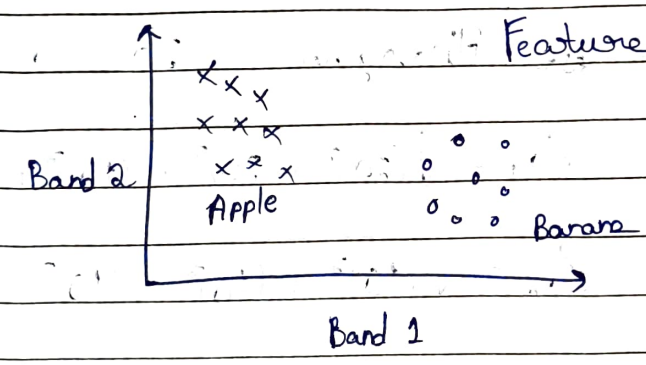
3. Homogeneity $= \sum_i \sum_j \frac{P(i,j)}{1+|i-j|}$

4. Correlation $= \sum_i \sum_j \frac{(i-\mu_i)(j-\mu_j)P(i,j)}{\sigma_i \sigma_j}$

→ Cluster + Mapping ⇒ Class

→ Match the input image to the best nearest available class

Support Vector Machine (SVM)



Linear SVM Eqn. $g(x) = W^T x + b$

$x = [R \;\; G \;\; B]^T$

Eg:-

Apple S.V. ($G = 70$) : $70 W_G + b = +1$

Banana S.V. ($G = 205$): $205 W_G + b = -1$

$\therefore 205 W_G - 70 W_G = -1 - 1$

$\therefore W_G = -0.014815$

$\therefore b = 2.037$

Final SVM model: $w = [0, -0.014185, 0]$

$b = 2.037$

Decision function:

$$f(x) = 0R - 0.014815G + 0B + 2.037$$

(R, G, B)

Prediction: $f(x) > 0 \rightarrow$ Apple

$f(x) < 0 \rightarrow$ Banana

Full margin between 2 supporting hyperplanes:

$$\text{Margin} = \frac{2}{\|w\|}$$

$137.5 - 70 = 67.5$

$205 - 137.5 = 67.5$

$\|w\| = 0.014815$

$$\Rightarrow \text{Margin} = \frac{2}{0.014815}$$

$= 135$. Answer